AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# TM354
# Software Engineering

## Final Examination
## Fall Semester 2024/2025

## Date: 8 /Jan./2025

AOU EGYPT

| | |
|---|---|
| **Number of Exam Pages:** 4 (including this cover sheet) | **Time Allowed:** 3 Hours |

## Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **Not** allowed.



CamScanner

## Part I – Multiple Choice Questions — 10 Marks

*Answer **all** the following multiple-choice questions. Each question is worth 2 Marks.*

1. Which of the following tests is used to make sure that the system can operate reliably at the limits of each of its resources.
   a. Restart test  b. Stress test  c. Security test  d. Acceptance test

2. Which of the following is **not** a property of a legacy system:
   a. Large in size  b. Critical to the business  c. Updated very few times  d. Lack documentation

3. Which property of a service indicates that services respond to each request as a 'one-off', without retaining any memory of previous requests:
   a. Statelessness  b. Location Transparency  c. Service description  d. Registry Entry

4. Which architecture is used where one program processes a stream of data and then passes the result to another process for further processing:
   a. Peer-to-peer  b. Data flow  c. Data-centered  d. Notification

5. The description of a set of operations a component makes available to other components is known as
   a. Provided interface  b. Required interface  c. Post condition  d. Interaction

## Part II – Short Answer Questions — 30 Marks

*Answer **all** the following short answer questions.*

1. Given the following piece of code, calculate the Cyclomatic complexity for the "run" method and the CBO (Coupling-between-objects) for the "Player" class (**IMPORTANT**: show how each value was calculated):

**[10 Marks]**

```
public class Player extends Thread {
   int portNumber;
   BufferedReader br = null;
   public Player (int portNumber, BufferedReader br) {
         this.portNumber = portNumber;
         this.br = br;
       }
   public void run() {
         try {
               ServerSocket serverSocket = new ServerSocket(portNumber)
               Socket clientSocket = serverSocket.accept();
               while (true) {
                     String tempStr = br.readLine();
                     if (tempStr != null) {
                           break;
                     }
               }
         }
         catch (Exception ex) {
         }
   }
}
```

AUC EGYPT



CamScanner

2. Briefly discuss 2 different levels of reuse other than Reuse of Software Components.

**[8 Marks]**

3. Consider that you have the below classes with some attributes mentioned. For each class suggest a class invariant based on the listed attributes only.

**[6 Marks]**

Class Publication (Attributes: numberOfCopies, loans)
Class CourseSection (Attributes: capacity, numberOfRegisteredStudents)

4. Consider a word-processing system similar to Microsoft Word, where one version of it can be installed on the user's computer and an alternative version could be provided as a cloud application with all the functionality provided through a web browser connected to the internet. Identify 2 non-functional requirement types that are directly related to the architecture to be used in the given example and explain how they are affected with respect to the version of the system being used.

**[6 Marks]**

## Part III – Problem Solving Questions — 60 Marks

*Answer **all** the following problem-solving questions. Each question is worth 20 Marks.*

1. Given the following database tables:

Room

| Number | RoomTypeID |
|---|---|
| 101 | 1 |
| 102 | 1 |
| 103 | 2 |

RoomType

| RoomTypeID | Kind | Cost |
|---|---|---|
| 1 | Single | 80 |
| 2 | Double | 100 |

a- Suggest a suitable section of class diagram with 2 classes that represent the 2 tables above with a suitable association and multiplicity values.
b- Provide an Object diagram to represent all the rows of the 2 tables

2. Develop a sequence diagram for the process of withdrawing money from an ATM with the following requirements:
   a- Create 4 objects aUserInterface, atm:ATMSystem, bank:Bank, account:Account.
   b- Create at least 6 messages as you find suitable with at least 1 self-message.
   c- Use the opt fragment once (Example: pin validation, checking for sufficient amount to withdraw, etc...).
   d- Use the loop fragment once (Example: requesting another transaction).

EGYPT



CamScanner

3. Consider an automatic parking barrier system with the following requirements:
   a. The barrier has 3 states (Closed, Open, Timed-Open)
   b. The states will change based on the following:
      i. The user can choose to change the state from Closed to Opened, where it will remain opened unless another action is selected.
      ii. The user can choose to change the state from either Opened or Closed to Timed-Open, where the barrier will start a timer of 2 minutes and will change automatically to Closed afterwards.

Draw the state machine diagram for the parking barrier system showing the 3 states and at least 5 transitions and their respective events using Guards as you find necessary.

---

**End of questions**

AOU EGYPT

CamScanner